

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 10

2024

1. Show that no iterate of the function $f(x) = \sin(x)$ is a contraction.
2. Consider the initial value problem

$$\frac{dy}{dx} = 2e^{-x} - y, \quad y(0) = 2.$$

- (a) Show that the contraction mapping theorem can be used to show that this problem has a unique solution in $C([0, 1])$, and find the first three approximations to this solution starting with $y_1(x) = 1$ for all $x \in [0, 1]$.
- (b) Solve the equation exactly. [Hint: It is a linear differential equation].
- (c) Compare your answers and explain what you find.

3. Use

- (a) Theorem 8.20 in your notes
- (b) Corollary 8.21 in your notes

to show that the initial value problem

$$\frac{dy}{dx} = (y + x^2)e^{x-1}, \quad y(1) = 4.$$

has a unique solution in $C([1, 3])$.

4. Recall that in $(\mathbb{R}, d_E)$ a subset is compact if and only if it is closed and bounded, but that this is not the case in general.
Consider the space $(\mathbb{Q}, d_E)$ with subset $A = [-\pi, \pi] \cap \mathbb{Q}$. Show that this also provides a counter-example to the claim that all closed, bounded subsets of a metric space are compact.
5. Suppose that A and B are compact subsets of a metric space X . Which of the following are necessarily compact? Give an appropriate proof or counter-example.
 - (a) $A \cup B$
 - (b) A'
 - (c) $A \cap B$
6. Let $f : X \rightarrow Y$ be a continuous function between metric spaces where X is a compact space.
 - (a) Prove that $f(\overline{A}) = \overline{f(A)}$ for all subsets A of X .
 - (b) Deduce that f is a closed function.